



Morphology II

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2026/09/14 (updated: 2026-09-12)

Please pick up your **name card** from the front

Assignment #1 will be published this Wednesday
(**Sept 16**)

It is due next Friday (**Sept 25**) at **11:59 pm**

You might not be able to solve all problems
before the class next Monday (**Sept 21**)

morpheme

smallest linguistic unit with meaning

Homophones and Allomorphs

homophones

- sound the same, but mean different things
 - *small-er* → -er = 'more' *help-er* → -er = 'agent'
 - *paper-s* → -s = PLURAL *see-s* → -s = 3SG PRESENT
- these are **different morphemes** that happen to sound the same
- **homophonous**

allomorphs

- mean the same thing, but are pronounced differently
 - *wug-s*, *peff-s*, *gutch-es* *talk-ed*, *buzz-ed*, *land-ed*
 - *magic-al*, *magic-ian* *clear-ing*, *clar-ify*
- each of these is a **separate morpheme** with multiple pronunciations and each pronunciation is called an **allomorph** of that morpheme

Types of Morphemes

Exercise I: Warm-up

compare the following set of words;

(a)	dog	board	laugh	teach
(b)	refute	refutable	irrefutable	
(c)	walks	walked	walking	
(d)	cranberry	raspberry	mulberry	
(e)	blackberry	blueberry	gooseberry	
(f)	workshop	snowboard	snowwhite	

1. what is the difference between (a) and other groups?
2. what is the difference between (b) and (c)?
3. what is the difference between (b, c) and (e)?
4. what is the difference between (d) and (e)?

Types of Morphemes I: Free vs. Bound

compare two **lexical entries** we have seen:

label	'dog'	'teach'
phonological representation	/dag/	/tɪ:tʃ/
semantic representation	{x: x is a dog }	{<x, y>: x teaches y}

label	'-er'	'-able'
phonological representation	/ər/	/əbəl/
semantic representation	one who does X action	able to be X -ed

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free morpheme

can stand alone as a word
e.g. *dog, cat, house,*

label	'-er'	'-able'
phonological representation	/ər/	/əbəl/
semantic representation	one who does X action	able to be X -ed

bound morpheme

cannot stand alone
e.g. *-er, -s, cran-*

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free morpheme

can stand alone as a word
e.g. *dog, cat, house,*

bound morpheme

cannot stand alone
e.g. *-er, -s, cran-*

ro-infix-ot

prefix-



-suffix

circum-

-fix

Types of Morphemes I: Free vs. Bound

free morpheme

can stand alone as a word

e.g. *dog*, *cat*, *house*,

bound morpheme

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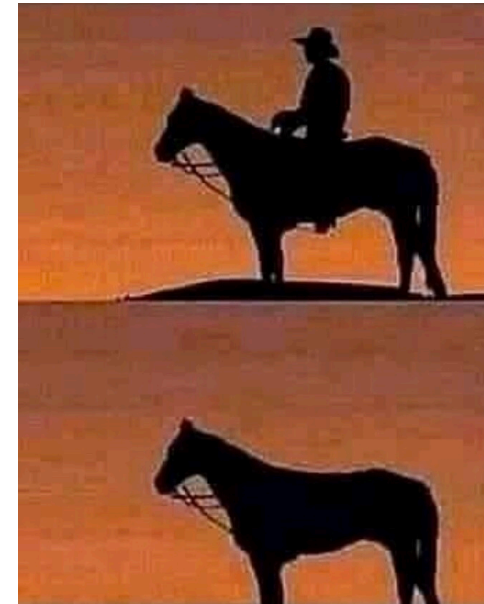
e.g. *-er*, *-s*, *cran-*



Exercise II: Bound Morphemes

how many **bound morphemes** do you spot?

humorless	cleverness	blamelessly
hopeless	carelessly	classroom
harmlessly	hopefully	fearlessness
discernment	mushroom	rapidly
stupidly	enrage	joyful
heiress	lioness	dumbness
encourage	senseless	stewardess
bathroom	assignment	enjoyment



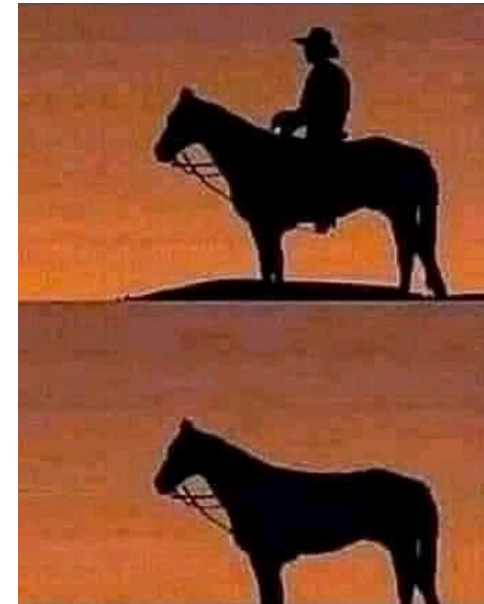
Cowboy

Cow

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humor less	clever ness	blame lessly
hope less	care lessly	classroom
harm lessly	hope fully	fear lessness
discern ment	mushroom	rapid ly
stupid ly	en rage	joy ful
heir ess	lion ess	dumb ness
en courage	sense less	steward ess
bathroom	assign ment	enjoy ment



Cowboy

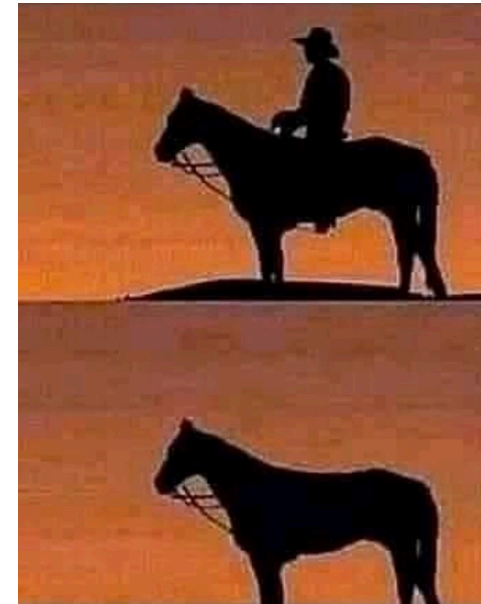
Cow

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- **-less, ly, -ment, --ess, en-, -ness, ful**
- about dis**cern**, con**cern**, re**cern**



Cowboy

Cow

1. why not **-lessness**?
2. why not **lio-ness**?
3. why not **-full**?

The Same Morphemes Might Behave Differently in Another Language

- a **single** morpheme in one language can be translated **into more than one** morpheme (or even multiple words) in another language
- just because a morpheme corresponds to a word in English **does NOT mean** it is necessarily a word in the given language

Spanish:

<i>habl-ar</i>	'to speak' (infinitive)
<i>habl-o</i>	'I speak'
<i>habl-as</i>	'you speak'
<i>habl-ábamos</i>	'we were speaking'
<i>habl-aríamos</i>	'If we spoke'
<i>habl-a</i>	'Speak!' (imperative)
<i>*habl</i>	???

- we can easily see that the morpheme **habl** corresponds to the English word **speak**, but it does not correspond to a word in Spanish, **habl** must always occur with other morphemes, so in Spanish it is a **bound morpheme**

Types of Morpheme II: Root vs. Affixes

root

- **"base"** morpheme to which other morphemes are attached
- contributes the **core meaning** of the word
- if a word has a single morpheme, that morpheme is a root
- examples: *danc-ing*, *dis-appear*, *help-less-ness*, *elephant*

affix

- morpheme that is **attached to the root**
- **prefixes** go before root, **suffixes** go after root – both are affixes
- a word can have multiple prefixes or suffixes
- examples: *in-complete*, *act-ive*, *un-trust-worthy*, *will-ing-ly*

Types of Morpheme II: Root vs. Affixes

root

- a word with **more than one root** is called a **compound** word:
 - *house-cat, ginger-bread, sun-flower*
- a word with a single **one root** is called **simple**, even if it has affixes:
 - *cat, eat-s, be-friend*

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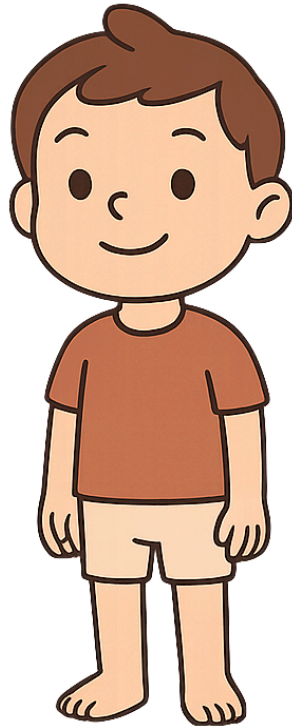
compound/simple vs. complex/simplex

Types of Morpheme II: Root vs. Affixes

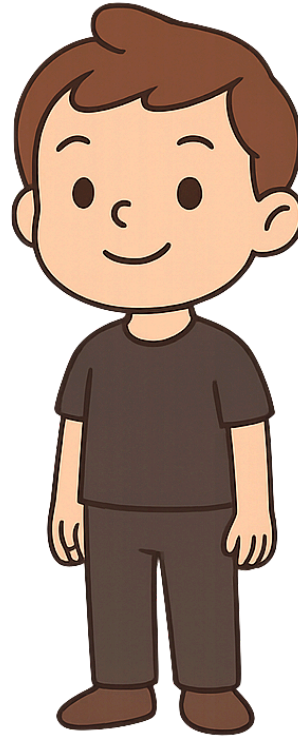
affix

- most common affixes are **prefixes** and **suffixes**
- some languages also exhibit **infixes**, which go **INSIDE** the root:
 - **Tagalog** *bili* 'buy' → *b-um-ili* 'to buy'
 - **English** *absolutely* → *abso-****-lutely*
- or **circumfixes**, which go **AROUND** the root:
 - **Indonesian** *ajar* 'teach' → *pel-ajar-an* 'lesson/teaching'
 - **English** *bold* → *em-bold-en*
- infixes and circumfixes are a bit rare, so if you think you have found one:
 - for an **infix**, check if it is really inside the root but not two pre/suffixes
 - for a **circumfix**, check if it is inseparable and means one thing

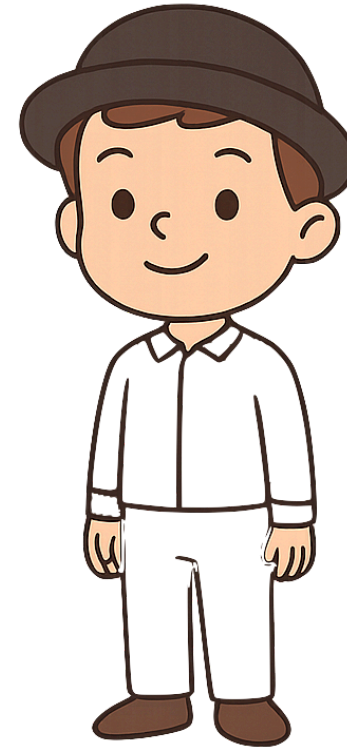
Dressing and Naming



Merlin

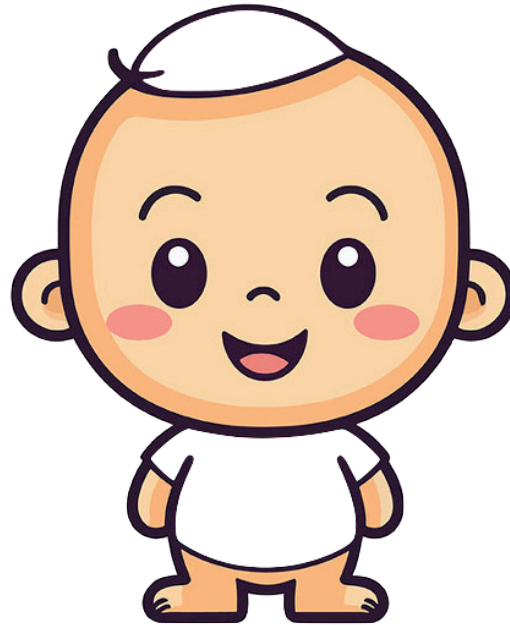


Merlin-Brown



Brown-Merlin-White

Dressing and Naming



and this is Arthur

Dressing and Naming



Dressing and Naming



Red-Arthur-Red

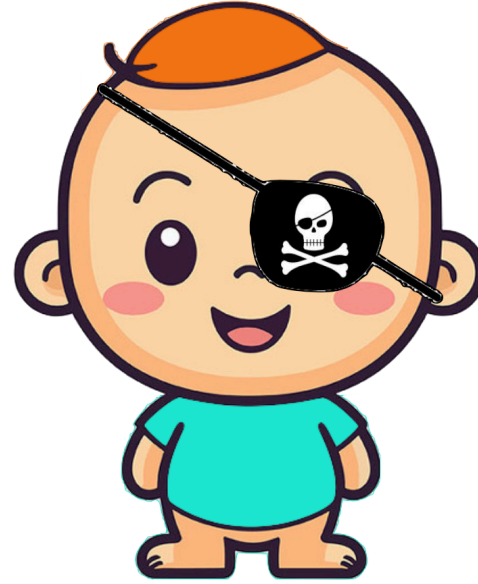


Green-Arthur-Red



Green-Arthur-Blue

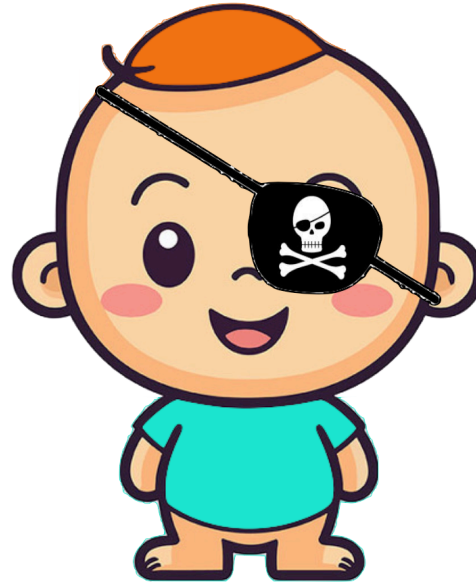
Dressing and Naming



Dressing and Naming



Green-BATMAN-Arthur-Blue



Orange-PIRATE-Arthur-Teal



Claret-PHANTOM-Arthur-Claret

Dressing and Naming



Dressing and Naming



BLUE-Arthur-HOODY



BLUE-Ar-EARRING-thur-HOODY

Inflectional Morphemes

violin
violin-s
violin-ist

agree
agree-d
agree-ment

large
larg-er
en-large

Inflectional Morphemes

violin
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large
larg-er
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consider these words and affixes on the left:

- the **first** affix in each group (marked **red**) doesn't really change the word's core meaning – instead, it adds grammatical nuance:
 - violin-s* = *violin* + **MORE THAN ONE**
 - agree-d* = *agree* + **PAST**
 - larg-er* = *large* + **MORE**
- the process of adding grammatical nuance is called **inflection**, and these affixes are therefore **inflectional morphemes**
 - inflecting a verb is **conjugating**
 - inflecting a noun is **declining**

Derivational Morphemes

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Derivational Morphemes

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consider these words and affixes on the left:

- the **second** affix in each group (marked **green**) changes the word's meaning substantially, and can even change its **part of speech** – adding them **makes a new word**
 - violin* (musical instrument, **N**)
*violin**ist*** (a person, **N**)
 - agree* (share opinions, **V**)
*agree**ment*** (contract, **N**)
 - large* (big, **Adj**)
***en**large* (make big, **V**)
- the process of forming new words is called **derivation**, and these affixes are therefore **derivational morphemes**

Types of Morphemes III

Inflectional vs. Derivational

inflectional

- **inflectional morphemes** are used to **inflect** words
- they add grammatical nuance but do not change the core meaning
- in English, they're mostly used to conjugate verbs (*jump* → *jump-ed*), pluralize nouns (*cat* → *cat-s*), and form the comparative and superlative of adjectives (*bright* → *bright-er* → *bright-est*)

derivational

- **derivational morphemes** are used to derive **new words**
- they change the core meaning of the word in some way
- they can, but do not always, change a word's part of speech
 - *simple* → *simpl-ify*, *exist* → *exist-ence*, *pack* → *un-pack*

Exercise III: Morphemes Types

for each word:

- indicate whether it is **mono-** or **poly**morphemic
- indicate whether it is **simple** or **compound**
- divide it into morphemes (use **hyphens**)

for each morpheme:

- indicate whether it is a **root**, **prefix**, or, **suffix**
- indicate whether it is a **free** or **bound** morpheme
- if it is an affix, indicate whether it is **inflectional** or **derivational**

1. unpacks

2. railroad

3. chameleon

4. backpacker

5. greater

6. finger

7. expectant

8. unbelievably

9. uncouth

10. meaningful

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1. unpacks

un-pack-s

2. railroad

rail-road

3. chameleon

chameleon

4. backpacker

back-pack-er

5. greater

great-er

6. finger

finger

7. expectant

expect-ant

8. unbelievably

un-believ-ab-ly

9. uncouth

un-couth

10. meaningful

mean-ing-ful

Summary for Different Morphemes

- **root** (base; **build**, **flower**)
- **affix** (morpheme that attaches to the base)
 - **prefix** (**un**-do)
 - **suffix** (walk-**ed**)
 - **infix** (rare in English)
 - **circumfix** (**em**-bold-**en**)
- **free vs. bound morpheme**
 - can appear as an independent word?
 - **blue**berry vs. **cran**berry
- **derivational morpheme** (piano --> pian-**ist**)
 - change core meaning even part of speech
- **inflectional morpheme** (jump --> jump-**ed**)
 - add some grammatical nuance
- **homophone** (fast-**er**, teach-**er**)
 - same sound, different meanings
- **allomorph** (**magic**, **magic**-ian)
 - same meaning, different sounds
- **simple word vs. compound word**
 - one root = simple (inter-**nation**-al)
 - more than one = compound (**sawhorse**)

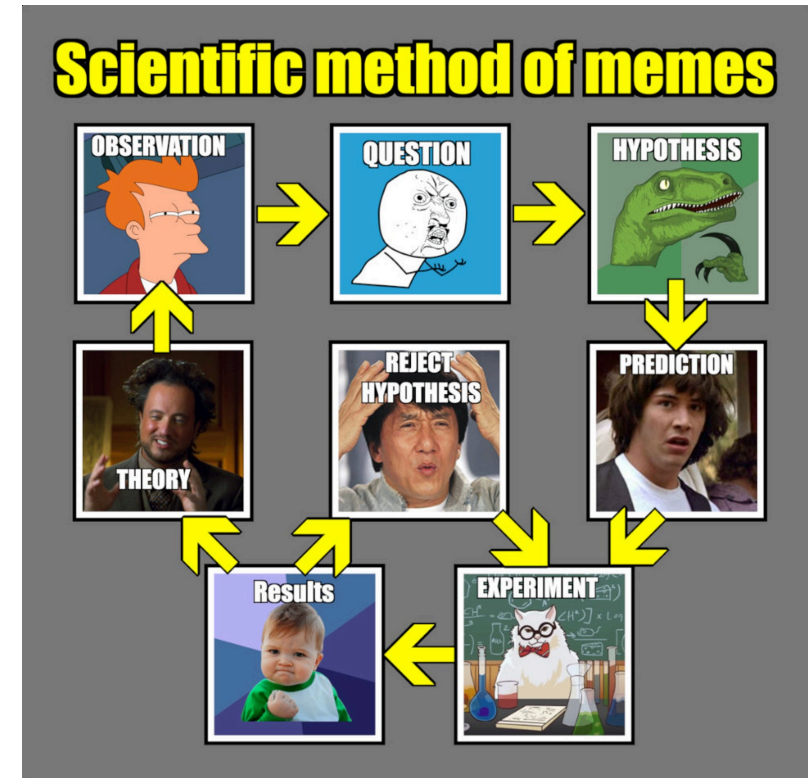
Morphological Analysis

Morphological Analysis

- **morphological analysis** is a technique for figuring out how the morphology of a language works.
- you can use it to learn how the morphology of an unfamiliar language works – as long as you have a set of words with different morphemes and their translations (**glosses**) into English.
- **steps:**
 1. identify any **repeated** sequences of sounds in the dataset – they may be a morpheme
 2. identify any **repeated** meanings in the English glosses
 3. try to match each meaning from step 2 with a potential morpheme from step 1

Notes for Morphological Analysis

1. try to find two words that differ by just one meaning or morpheme)
2. look out for any potential **homophones** or **allomorphs**
3. don't be afraid to **revise your hypotheses** if the data suggests you're on the wrong track
4. **order** of morphemes matters
5. **affixes are picky**



Excercise V: Morphological Analysis

Isthmus Zapotec, Oto-Manguan

01. palu	'stick'	06. spalube	'his stick'	11. spalulu	'your stick'
02. kuba	'dough'	07. skubabe	'his dough'	12. skubalu	'your dough'
03. geta	'tortilla'	08. sketabe	'his tortilla'	13. sketalu	'your tortilla'
04. bere	'chicken'	09. sperebe	'his chicken'	14. sperelu	'your chicken'
05. dolo	'rope'	10. sto?obe	'his rope'	15. sto?olu	'your rope'

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- what morphemes does Isthmus Zapotec use to indicate the following concepts?
 - stick, dough, tortilla, chicken, rope
 - third person singular (s/he)
 - second person plural (plural 'you')
 - possession ('s in English)
- the word for 'four' is *tapa*, then how would we say 'his four' and 'your four'?

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BONUS QUESTION

if **spalube** is grammatical
but neither * **s**palu nor * palu**e** is grammatical
what does this tell us about morpheme *s-* and *-be*?

(This might not be the actual case and it is only an assumption)

Excercise VI: Morphological Analysis

Luiseno, Uto-Aztecan

a.	nokaamaj	'my son'	m.	pokaamaj	'his son'
b.	7oki	'your house'	n.	poki	'his house'
c.	potaana	'his blanket'	o.	notaana	'my blanket'
d.	7ohuukapi	'your pipe'	p.	pohuukapi	'his pipe'
e.	7otaana	'your blanket'	q.	nohuukapi	'my pipe'
f.	noki	'my house'	r.	7okaamaj	'your son'
g.	7omkim	'your (pl.) houses'	s.	pompeewum	'their wives'
h.	nokaamajum	'my sons'	t.	pomki	'their house'
i.	popeew	'his wife'	u.	tfampeewum	'our wives'
j.	7opeew	'your wife'	v.	tfamhuukapim	'our pipes'
k.	7omtaana	'your (pl.) blanket'	w.	7omtaanam	'your (pl.) blankets'
i.	tfamhuukapi	'our pipe'	x.	pomkaamaj	'their son'

give the luiseno morpheme that corresponds to each English translation

- | | | |
|--------------|-----------------|--------------------|
| a. 'son' | e. 'my' | i. 'their' |
| b. 'house' | f. 'his' | j. (plural marker) |
| c. 'blanket' | g. 'your (sg.)' | k. 'pipe' |
| d. 'wife' | h. 'your (pl.)' | l. 'our' |

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Luiseño, Uto-Aztecan

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b.	ʔoki	'your house'	n.	poki	'his house'
c.	potaana	'his blanket'	o.	notaana	'my blanket'
d.	ʔohuukapi	'your pipe'	p.	pohuukapi	'his pipe'
e.	ʔotaana	'your blanket'	q.	nohuukapi	'my pipe'
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a.	no kaamaj	'my son'
h.	no kaamaj um	'my sons'
m.	po kaamaj	'his son'
r.	ʔo kaamaj	'your son'
x.	po kaamaj	'their son'

b.	ʔ oki	'your house'
d.	ʔ ohuukapi	'your pipe'
e.	ʔ otaana	'your blanket'
j.	ʔ opeew	'your wife'
r.	ʔ okaamaj	'your son'

Excercise VI: Morphological Analysis

Luiseño, Uto-Aztecan

- a. no**kaamaj** 'my son'
- h. no**kaamaj**um 'my sons'
- m. po**kaamaj** 'his son'
- r. 7o**kaamaj** 'your son'
- x. po**kaamaj** 'their son'

- b. 7o**k**i 'your house'
- d. 7o**k**huukapi 'your pipe'
- e. 7o**k**taana 'your blanket'
- j. 7o**k**opeew 'your wife'
- r. 7o**k**kaamaj 'your son'

Excercise VI: Morphological Analysis

Luiseno, Uto-Aztecan

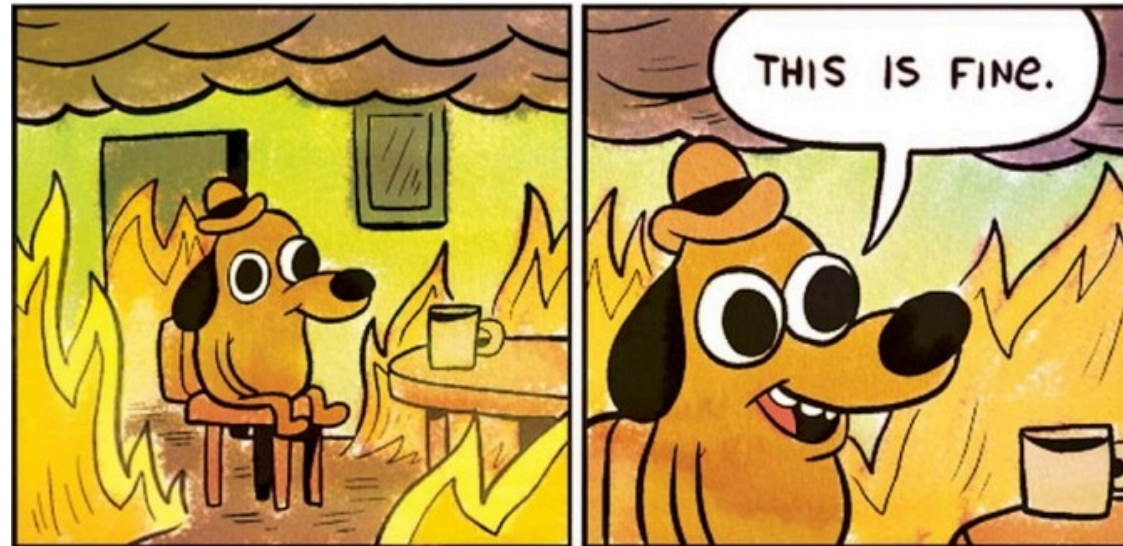
a.	no kaamaj	'my son'	b.	7o ki	'your house'
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m.	po kaamaj	'his son'	e.	7o taana	'your blanket'
r.	7o kaamaj	'your son'	j.	7o peew	'your wife'
x.	po kaamaj	'their son'	r.	7o kaamaj	'your son'

give the luiseno morpheme that corresponds to each English translation

a.	'son'	kaamaj	e.	'my'	no-	i.	'their'	pom-
b.	'house'	ki	f.	'his'	po-	j.	(plural marker)	-um / -m
c.	'blanket'	taana	g.	'your (sg.)'	7o-	k.	'pipe'	huukapi
d.	'wife'	peew	h.	'your (pl.)'	7om-	l.	'our'	tfam-

no homework is due this week

readings Sections 4.3, 4.4 and 4.5 of Chapter 4: Morphology from *Language Files*



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